

# CE902 (2022) Worked Solutions

## Professional Practice and Research Methodology

**Scope note.** The uploaded paper explicitly states that **Section A is unavailable**. Therefore this document solves **all questions visible in the source file**: Section B, Questions 18–21. The aim is not to provide a bare answer key, but to show the reasoning, formulas and calculations that would justify each answer in an exam script.

All monetary calculations below use the present value relation stated in the exam paper:  $PV = X / (1 + r)^t$ . Intermediate figures are kept to full calculator precision and then sensibly rounded in the tables.

Discounting starts from the investment at year 0, then each inflow is divided by  $(1.08)^t$ .



## Question 18 – Project appraisal using discounting and NPV

The exam gives projected cash inflows for Projects A, B and C, together with an 8% interest rate. The correct appraisal procedure is: first compute the *discount factor* for each year, then convert each future cash inflow into a present value, and finally compare the projects using *Net Present Value* (NPV), i.e. discounted inflows minus the initial capital cost.

### Step 1: Discount factors

Using  $DF_t = 1 / (1 + r)^t$  with  $r = 0.08$ :

Year 1:  $DF = 1 / (1.08)^1 = 0.925926$

Year 2:  $DF = 1 / (1.08)^2 = 0.857339$

Year 3:  $DF = 1 / (1.08)^3 = 0.793832$

Year 4:  $DF = 1 / (1.08)^4 = 0.735030$

Year 5:  $DF = 1 / (1.08)^5 = 0.680583$

| Year | Formula        | Discount factor |
|------|----------------|-----------------|
| 1    | $1 / (1.08)^1$ | 0.9259          |
| 2    | $1 / (1.08)^2$ | 0.8573          |
| 3    | $1 / (1.08)^3$ | 0.7938          |
| 4    | $1 / (1.08)^4$ | 0.7350          |
| 5    | $1 / (1.08)^5$ | 0.6806          |

### Step 2: Discount the cash inflows for each project

For each year, the discounted cash inflow is the stated inflow multiplied by the relevant discount factor. Because the data are already in **£000**, the table below keeps the same unit throughout.

| Year | DF     | A inflow | A discounted | B inflow | B discounted | C inflow | C discounted |
|------|--------|----------|--------------|----------|--------------|----------|--------------|
| 1    | 0.9259 | 75       | 69.44        | 100      | 92.59        | 50       | 46.30        |
| 2    | 0.8573 | 125      | 107.17       | 200      | 171.47       | 75       | 64.30        |

| Year                           | DF     | A inflow | A discounted  | B inflow | B discounted  | C inflow | C discounted  |
|--------------------------------|--------|----------|---------------|----------|---------------|----------|---------------|
| 3                              | 0.7938 | 125      | 99.23         | 300      | 238.15        | 250      | 198.46        |
| 4                              | 0.7350 | 100      | 73.50         | 300      | 220.51        | 300      | 220.51        |
| 5                              | 0.6806 | 75       | 51.04         | 150      | 102.09        | 200      | 136.12        |
| <b>Total discounted inflow</b> |        |          | <b>400.39</b> |          | <b>824.81</b> |          | <b>665.68</b> |

### Step 3: Compute NPV for each project

The definition used is  $NPV = \text{Total discounted inflow} - \text{initial capital cost}$ .

Project A:  $NPV = 400.39 - 300.00 = \mathbf{100.39 \text{ (£000)}}$

Project B:  $NPV = 824.81 - 500.00 = \mathbf{324.81 \text{ (£000)}}$

Project C:  $NPV = 665.68 - 300.00 = \mathbf{365.68 \text{ (£000)}}$

Project **C** is the most financially attractive because it has the **highest positive NPV**. A positive NPV means the project more than covers the cost of capital at 8%; the largest positive NPV is therefore the best choice under the assumptions given.

## Question 19 – Company structure and gearing

### Part (a): five differences between a limited company and a sole trader

| Aspect                        | Limited company                                    | Sole trader                                                                  |
|-------------------------------|----------------------------------------------------|------------------------------------------------------------------------------|
| Legal status                  | Separate legal entity from its owners              | No separate legal identity; the owner and business are the same legal person |
| Liability                     | Owners normally have limited liability             | Owner has unlimited liability                                                |
| Ownership                     | Usually owned by shareholders                      | Owned by one individual                                                      |
| Continuity                    | Can continue despite changes in ownership          | Usually ends if the owner dies or stops trading                              |
| Administration and regulation | More filing, reporting and compliance requirements | Simpler administration and fewer formal obligations                          |

### Part (b): meaning of gearing and examples of ratios

A company's **gearing** (or leverage) describes the extent to which the business is financed by **borrowed funds** rather than by owners' equity. The higher the debt component, the higher the gearing.

- Debt-to-equity ratio = Total debt / Shareholders' equity
- Debt-to-capital ratio = Total debt / (Total debt + Equity)
- Interest cover ratio = Profit before interest and tax (PBIT or EBIT) / Interest payable

### Part (c): why high gearing is riskier

A highly geared firm must meet relatively large fixed commitments such as interest and capital repayments. If profits or cash flow fall, those fixed payments do not disappear. That makes the business more exposed to insolvency risk, more sensitive to interest-rate changes, and less flexible when trying to raise further finance. In short: debt magnifies both potential returns *and* potential financial distress.

## Question 20 – Referencing, summarising and plagiarism

### Part (a): does the statement about Tim Berners-Lee require a reference?

In most academic contexts, **no reference is normally required** for the statement “Tim Berners-Lee invented the World Wide Web” because it is a widely known, non-controversial historical fact and would usually be treated as **common knowledge**. A reference would become appropriate if the assignment needed a precise historical source, a disputed interpretation, or the exact wording from a particular author.

### Part (b): what belongs in a summary?

| Item                           | Include?   | Reason                                                                                                                     |
|--------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------|
| A. Key points from the article | <b>Yes</b> | A summary should condense the article's main ideas.                                                                        |
| B. The author's viewpoint      | <b>Yes</b> | A competent summary should reflect the author's argument or stance.                                                        |
| C. Your viewpoint on the topic | <b>No</b>  | That belongs in evaluation or discussion, not in a neutral summary.                                                        |
| D. In-text referencing notes   | <b>Yes</b> | The summary still needs clear source attribution; page numbers are essential for direct quotation and useful in note form. |
| E. Your personal opinion       | <b>No</b>  | Personal judgement is separate from summarising the source faithfully.                                                     |

### Part (c): comment on the student extract

Yes — the student extract **does amount to plagiarism**. Although some words have been changed, the passage keeps the same sequence of ideas, the same specialist abbreviations (FDI, ITT, MNEs, IPR, IPP), and closely follows the structure and phrasing of the original source. This is often called **patchwriting**: superficial rewording without proper acknowledgement. To avoid plagiarism, the student would need to (1) cite Yang and Clarke, (2) genuinely restate the ideas in an original sentence structure, and (3) use quotation marks if any distinctive wording is retained.

## Question 21 – STAR technique and covering letters

### Part (a): matching the STAR categories

| STAR category | Matching statement | Why it fits                                                                          |
|---------------|--------------------|--------------------------------------------------------------------------------------|
| Situation     | Statement 3        | It sets the background context: a departmental consultation on proposed legislation. |
| Task          | Statement 1        | It states the responsibility assigned to the applicant.                              |
| Action        | Statement 4        | It explains the steps actually taken by the applicant.                               |
| Result        | Statement 2        | It describes the outcome and the positive impact of the work.                        |

### Part (b): classify the covering-letter statements

| Category                      | Letters    | Reasoning                                                                                 |
|-------------------------------|------------|-------------------------------------------------------------------------------------------|
| Opening statements            | F, L       | These introduce the purpose of the letter and identify the advertisement/reference.       |
| Experience and qualifications | A, B, C, G | These describe current role, responsibilities, prior experience and evidence from the CV. |
| Reasons for applying          | D, E       | These explain motivation and why the applicant wants the role.                            |
| Closing statements            | H, I, J, K | These signal availability, willingness to discuss further and polite closure.             |

This completes the worked solutions for every question visible in the uploaded CE902 (2022) paper. Because Section A is unavailable in the source file, no reliable solution for those missing multiple-choice questions can be produced without the actual question text.